1 Overview

When a ray hits a surface, at each surface hit the ray tracer blends direct lighting, perfect-mirror reflection, and refraction using Fresnel-based weights and material parameters. The recursion terminates when the maximum depth is reached or the material contributes no reflective/transmissive energy.

This short tutorial explains how the ray tracing mathematics is applied in the context of my implementation.

2 Recursive Equation

Let C_d be the RGB color returned at recursion depth d for a ray that hits a surface; D be the local (direct) lighting at the hit point (diffuse + specular + ambient, aka Phong model), and $\mathcal{T}$ the (optional) color tint operator.

$$C_d \;=\; w_d D \;+\; w_r C_{d-1}^{\text{refl}}(n_1) \;+\; w_t \mathcal{T}\big(C_{d-1}^{\text{refr}}(n_2), \text{tint}\big) \tag{2.1}$$

Base case (no recursion or negligible reflective/transmissive energy):

$$C_d \;=\; (1-k_t)D \tag{2.2}$$

The following coefficients appear in Eq. (2.2):

- $0 \le k_r \le 1$: material reflectiveness
- $0 \le k_t \le 1$: material transparency
- R the Schlick coefficient, which approximates the reflectance of a refractive medium given the incident angle θ_i
- $0 \le w_r \le 1$: reflection weight
- $0 \le w_t \le 1$: transmitance (refraction) weight
- $0 \le w_d \le 1$: direct light weight, such that $w_d + w_r + w_t = 1$

$$\text{tint}_w = \texttt{material.tint} \cdot k_t, \qquad c' = \min\!\left(255,\, c \cdot \left[(1-\text{tint}_w) + \text{tint}_w \cdot \frac{c_{\text{curr}}}{255}\right]\right)$$

Fresnel reflectance approximation via Schlick:

$$R \;=\; r_0 + (1-r_0)(1-\cos_i)^5, \qquad r_0 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$$

where $\cos_i = -\hat{n} \cdot \hat{\imath}$, with $\hat{n}$ the oriented surface normal (pointing out of the refracted medium) and $\hat{\imath}$ the incident direction.

Weights with respect to R:

$$w_t = k_t(1-R) \quad \text{(refraction)} \tag{2.3}$$

$$w_r = k_r + k_t R \quad \text{(reflection)} \tag{2.4}$$

$$w_d = 1 - \min\big(1, w_r + w_t\big) \quad \text{(direct residual)} \tag{2.5}$$

Note: in the implementation, D is pre-scaled by $(1-k_t)$ before mixing to account for energy loss due to transparency.

Sometimes, Total Internal Reflection (TIR) may occur

$$\text{if refraction fails:} \quad w_t \leftarrow 0, \qquad w_r \leftarrow \min(1, w_r + k_t)$$

In this case, no refracted child is spawned (see next session).

3 Child Rays and Depth

Let $\hat{\imath}$ be the incident ray, $\hat{n}$ the normal at the intersection pointing away from the refraction medium. Then the ray will be split between the reflection child (away from the medium) and the refraction child (towards the medium).

Reflection child:

$$\hat{r} = \hat{\imath}\,\text{reflect about } \hat{n}$$

$$\text{origin} = p_{\text{hit}} + \hat{n}\,\epsilon$$

$$C_{d-1}^{\text{refl}}(n_1) = \texttt{TraceRay}\big(\text{ray}(\text{origin}, \hat{r}), d-1, n_1\big)$$

Refraction child (if not TIR):

$$\hat{t} = \text{Snell}(\hat{\imath}, \hat{n}, \eta), \quad \eta = \frac{n_1}{n_2}$$

$$\text{origin} = p_{\text{hit}} + \hat{t}\,\epsilon$$

$$C_{d-1}^{\text{refr}}(n_2) = \texttt{TraceRay}\big(\text{ray}(\text{origin}, \hat{t}), d-1, n_2\big)$$

Mediums n_1, n_2 and η are set by entry/exit orientation:

$$\texttt{RayOrientation} \Rightarrow (n_1, n_2, \eta, \cos_i), \quad n_{\text{surround}} = \texttt{SurroundingIOR}(\cdot)$$

4 Miss (No Intersection)

If the ray misses all objects:

$$C_d \;=\; \text{BG}(\hat{\imath}) \;=\; \texttt{SampleBackground}(\hat{\imath})$$

5 Implementation Mapping

The final combination in code matches Eq. Eq. (2.1):

$$\texttt{ret.color} = D \cdot w_d \;+\; \texttt{refl\_col} \cdot w_r \;+\; \texttt{refr\_color} \cdot w_t$$

with:

$$D = \texttt{lights\_.ColorAt(...)} \cdot (1 - k_t), \quad w_t = \texttt{trans\_weight}, \quad w_r = \texttt{refl\_weight}, \quad w_d = 1 - \min(1, w_r + w_t)$$